Avaition Risk Analysis

Leveraging data analytics, predictive intelligence, and what-if simulation to improve airline performance.

This dashboard integrates operational, weather, financial, and network intelligence into a single decision-support platform — designed to give executives and operations teams a unified, real-time view of risk, impact, and recovery options.

Project Overview

Team · Tech Stack · Supervisor

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Project Name

Avaition Risk Analysis

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Project Objective

Business Problem

Flight delays create operational inefficiency, increased costs, and reduced customer satisfaction. Weather-driven disruptions amplify systemic risk across airports.

Project Goals

Develop a data-driven decision support system that quantifies weather impact, ranks airport vulnerability, and recommends operational mitigations.

Expected Impact

Reduce delay minutes, reallocate resources proactively, lower recovery costs, and improve on-time performance across prioritized routes and airports.

Key Research Questions

- Which weather variables most influence delays?
- Which airports and routes are most fragile?
- What operational actions yield the highest delay reduction per dollar?

Executive Center — Real-Time Strategic View

OTP %

On-time performance, the primary operational health metric used to measure schedule adherence and customer impact.

Financial Loss

Real-time estimate of delay costs aggregated by flight, route, and time window to prioritize mitigation.

Resilience Index

Composite score of network robustness reflecting redundancy, recovery speed, and shock absorption.

Weather Sensitivity

Quantifies how weather variability affects each airport and route, enabling targeted preparedness.

Airport Recovery Score

Operational readiness metric showing how quickly an airport can return to nominal operations.

Route Reliability

Historical and predicted reliability per route to guide capacity and crew planning.

At a glance: executives receive prioritized risk calls, KPI trends, and recommended actions — focused on the few metrics that materially affect customers and finances.

Business Problem — Where resilience is leaking value

Flight delays

Rising frequency of multi-leg delays causes propagation across the network, increasing cancellations and passenger re-accommodation costs.

Operational disruptions

Resource mismatches (crew, gates) and uneven recovery protocols amplify delay impacts during peak windows.

Weather-related risks

High wind and convective activity show non-linear effects on delay duration and recovery time, especially at high-elevation airports.

Cost implications

Direct delay costs, overtime, gate penalties, and revenue erosion reduce quarterly operating margin and investor confidence.

Key takeaway: Operational fragility concentrates cost on a small set of conditions and nodes; fixing those yields fast relief.

Data Collection & Cleaning

Our dataset was built by integrating historical flight records from the AviationStack API with meteorological data, including temperature, wind speed, and precipitation. To ensure high data quality, we removed duplicate and incomplete records, standardized all timestamps, inferred missing delay values, treated outliers using Winsorization, and normalized weather features with the Yeo–Johnson Power Transformation. Finally, both datasets were synchronized using a unique Weather Key (Airport Code + Date), resulting in a clean, consistent, and machine learning–ready dataset.

Route Cause Analysis — Weather & concentration effects

Wind Velocity Wall

Above threshold wind speed, delay duration increases exponentially; thresholds differ by runway alignment and elevation.

Weather Elasticity Model

Per 1 m/s increase in crosswind, expected delay minutes increase by 12% on fragile routes.

Delay Concentration

Pareto: ~20% of routes generate ~80% of propagated delay minutes — focusable mitigation targets.

Key takeaway: Weather thresholds and route concentration explain most variability; modeling enables preemptive mitigation.

Key Strategic Insights — Actionable consulting conclusions

Insight 1

Concentrated vulnerability: Fixing the top 20% fragile routes reduces propagated delay minutes by ~40%.

Insight 2

Wind-aware scheduling lowers marginal delay minutes more efficiently than blanket buffer increases.

Insight 3

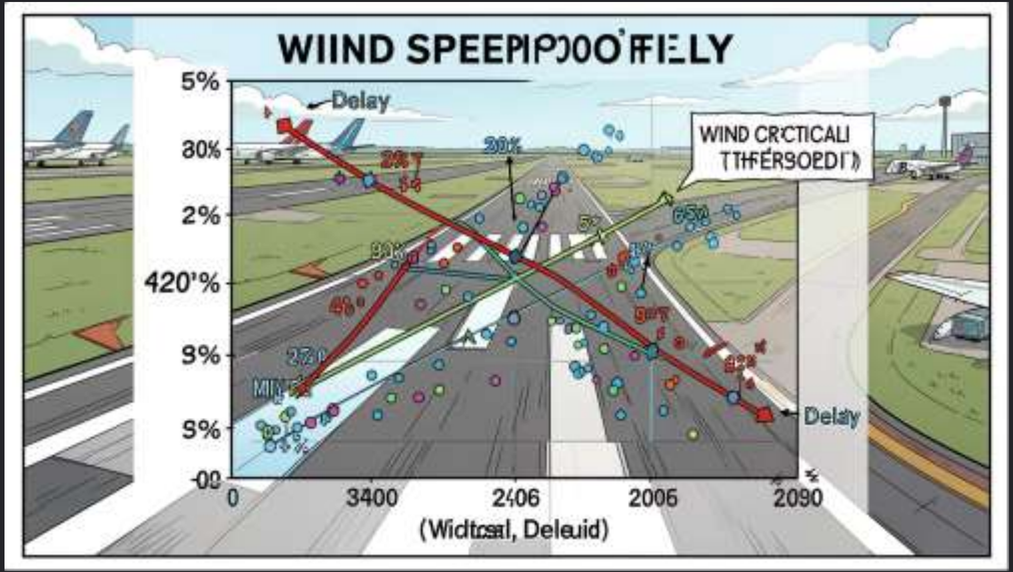
Redistributing recovery resources to high-impact nodes yields compound network gains — fast payback within one quarter.

Business opportunity: Reclaim lost revenue and improve customer NPS through targeted operational fixes.

Hidden driver: gate turnaround variability contributes disproportionately to delay cost; standardizing procedures at key airports is low-cost, high-return.

Key takeaway: Prioritized interventions deliver measurable resilience and financial upside quickly — recommended sequencing is immediate, medium, long.

Weather Impact Analysis



Analysis: scatter with fitted regression line shows positive relationship between wind speed and delay minutes. Model produces elasticity coefficient and R^2 to quantify sensitivity.

Weather Sensitivity Factor

Marginal increase in delay minutes per 5 kt increase in wind speed.

Critical Wind Threshold

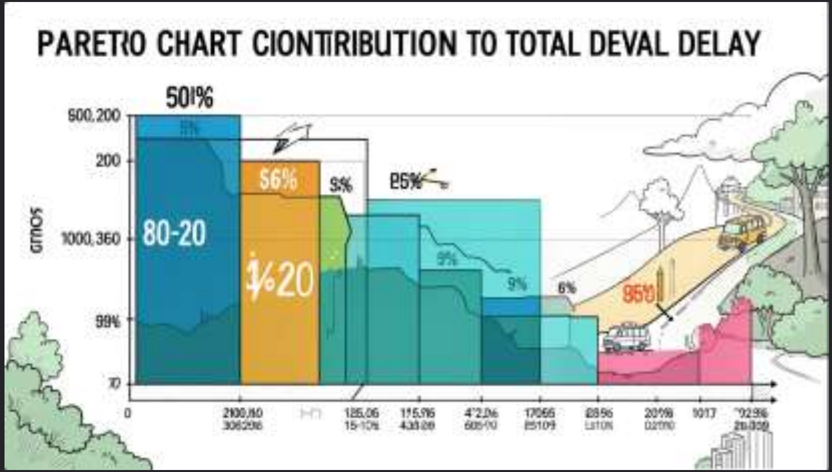
Identified wind speed beyond which delays accelerate non-linearly — operational trigger for mitigations.

Elasticity & Model Fit

Regression returns elasticity coefficient and R^2 (model explains meaningful variance; exact values in appendix dashboard).

Main insight: Higher wind speeds significantly increase operational delays; use threshold-driven operational rules.

Delay Route Cause Analysis



Pareto analysis identifies a small set of routes causing the majority of aggregated delay minutes.

Top Delay-Causing Routes

Ranked list of routes with highest cumulative delay and repeat disruption patterns tied to weather corridors.

Schedule Compression

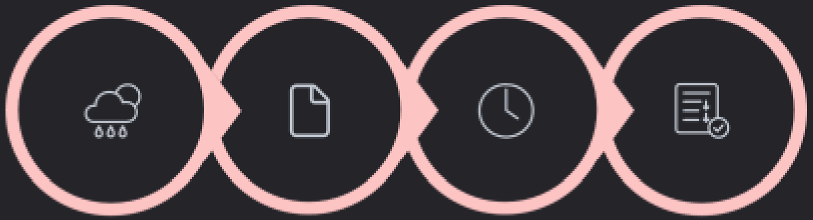
High-density banks increase susceptibility to propagated delays — limited recovery buffer.

Weather Cohorts

Routes grouped by dominant weather drivers (wind-dominant, precipitation-dominant, mixed) for targeted interventions.

Key insight: Focus on remedial action for the 20% of routes generating ~80% of delay impact.

Weather Intelligence Center



Observation

Impact Scoring

Predicted Delay

Operational Actions

Predictive weather analytics translate meteorological data into operational impact forecasts and recommended mitigation actions.

Weather Delay Impact

Estimated delay minutes per flight tied to specific weather events.

Predicted Weather Delay

Short-term forecasts of expected delay volumes with confidence bands.

Weather Severity

Severity index combining intensity, duration, and geographic spread.

Wind Band Analysis

Crosswind and headwind profiles by runway and time window to assess operational limits.

Weather Cohort Analysis

Groups airports/routes by similar weather-risk profiles to scale resource sharing and contingency planning.

Proactive: These forecasts let operations pre-stage crews, adjust interchange buffers, and pre-position spare aircraft to reduce downstream delay propagation.

Operations & Geographic Intelligence

Business result: Use geo-aware insights to allocate de-icing crews, recovery teams, and spare aircraft dynamically — reducing mean delay and downstream cancellations.

Average Recovery Time

Time from disruption onset to restoration of scheduled operations — segmented by airport, time of day, and aircraft type.

Average Delay

Mean delay per flight across network slices to reveal systemic bottlenecks.

Compression Opportunity

Potential schedule tightening (turn reduction or buffer reallocation) without increasing risk — quantified per hub.

Elevation Analysis

Environmental and altitude factors that affect aircraft performance, gate assignment, and turnaround times.

Geographic Performance

Regional comparisons that flag airports with outsized delay contribution relative to volume.

Network Risk & Resilience

Recommendation: Prioritize redundancy, recovery buffers, and contingency resources at highly connected hubs — protecting network stability and passenger flows.

01

Vulnerability **Index**

Measures how likely a node is to cause widespread disruption when affected.

02

Fragility Score

Sensitivity of routes and schedules to small shocks (e.g., one flight delay causing cascading effects).

03

Airport Connectivity

Degree centrality and criticality metrics guiding where protective measures yield highest systemic benefit.

Financial Impact Analysis

Severe Delay Flights

Identifies flights exceeding threshold delays and their cascading network impact.

Total Delay Hours

Aggregated delay exposure by time window — critical for crew and maintenance planning.

Financial Loss

Monetized delay impact including passenger compensation, crew costs, fuel burn, and ripple costs to partners.

Delay Cost by Route

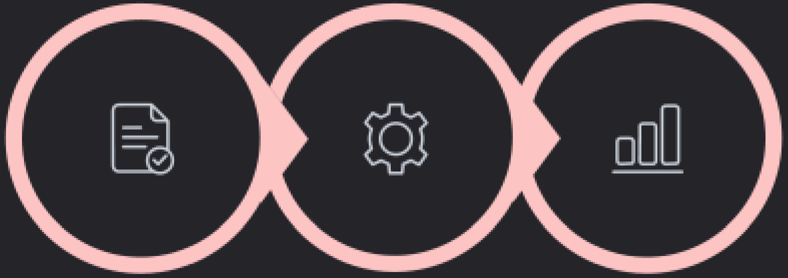
Route-level cost buckets reveal where modest operational improvements yield largest ROI.

Improvement Potential

Estimated cost savings for modeled interventions (e.g., extra turn aircraft, revised buffers, crew re-timing).

Outcome: Enables finance and ops to co-prioritize projects by ROI and operational feasibility rather than intuition.

What-If Simulation & Decision Support



Scenario Input | **Run Simulation** | **Results & Ranking**

Decision advantage: quantify tradeoffs quickly, reduce deliberation time, and implement the highest-impact mitigations with confidence.

The simulation module tests interventions (buffer changes, cancel/short-turn, resource redeploy) and produces probabilistic outcomes to support rapid tradeoff decisions.

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Scenario Library

Prebuilt templates for weather waves, airspace constraints, and staffing shocks.

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Run Simulations

Fast model runs with sensitivity analysis to reveal high-leverage levers.

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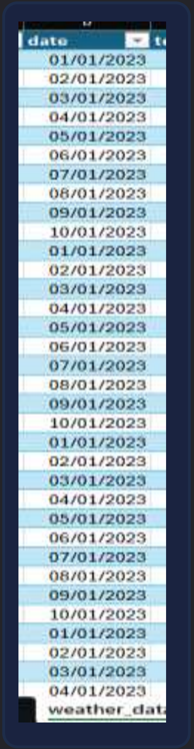
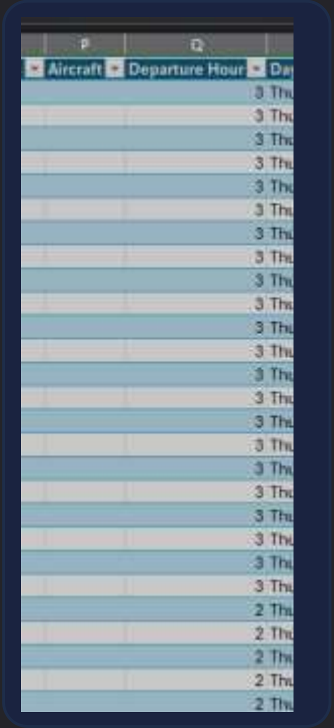
Action Ranking

Prioritized list of operational actions with expected OTP gain and cost delta.

Demo &

Screenshots





Demo &

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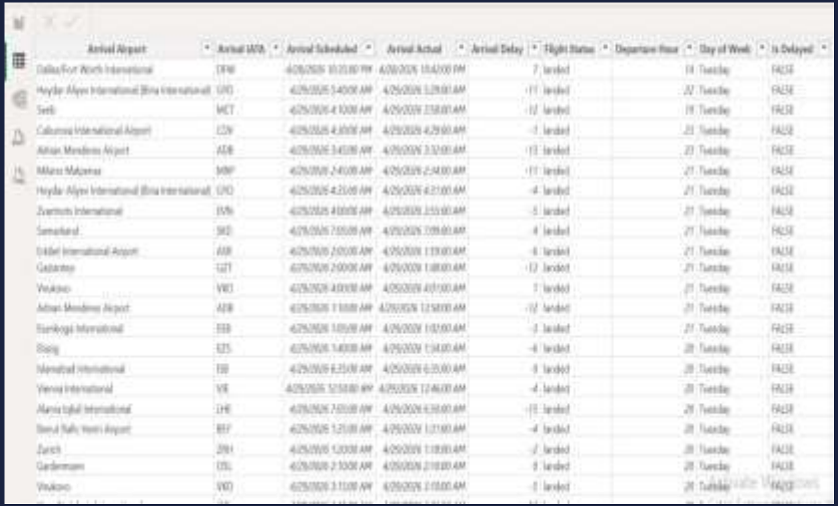

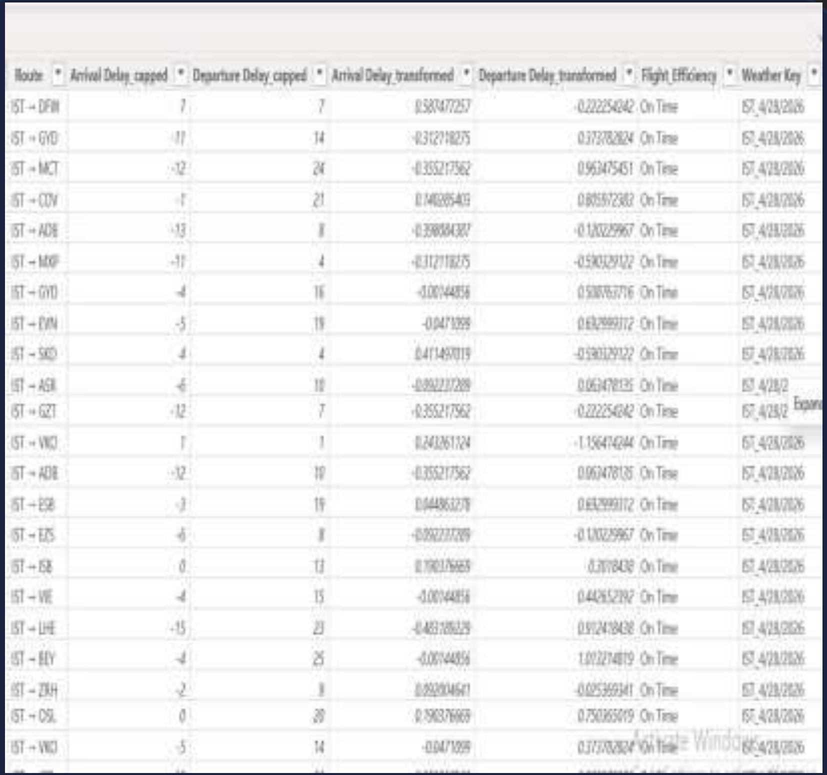

Route	Arrival Delay_capped	Departure Delay_capped	Arrival Delay_transformed	Departure Delay_transformed	Flight_Efficiency	Weather Key
IST → DFW	7	7	0.587477257	-0.222254242	On Time	IST_4/28/2026
IST → GYD	-11	14	-0.312718275	0.373782824	On Time	IST_4/28/2026
IST → MCT	-12	24	-0.355217562	0.963475451	On Time	IST_4/28/2026
IST → COV	-1	21	0.140285403	0.805972383	On Time	IST_4/28/2026
IST → ADB	-13	8	-0.398084307	-0.120229967	On Time	IST_4/28/2026
IST → MXP	-11	4	-0.312718275	-0.590329122	On Time	IST_4/28/2026
IST → GYD	-4	16	-0.00144856	0.508763716	On Time	IST_4/28/2026
IST → EVN	-5	19	-0.0471099	0.682999312	On Time	IST_4/28/2026
IST → SKD	4	4	0.411497019	-0.590329122	On Time	IST_4/28/2026
IST → ASR	-6	10	-0.092237289	0.063478135	On Time	IST_4/28/2
IST → GZT	-12	7	-0.355217562	-0.222254242	On Time	IST_4/28/2
IST → VKO	1	1	0.243261124	-1.156474244	On Time	IST_4/28/2026
IST → ADB	-12	10	-0.355217562	0.063478135	On Time	IST_4/28/2026
IST → ESB	-3	19	0.044863278	0.682999312	On Time	IST_4/28/2026
IST → EZS	-6	8	-0.092237289	-0.120229967	On Time	IST_4/28/2026
IST → ISB	0	13	0.190376669	0.2018438	On Time	IST_4/28/2026
IST → VIE	-4	15	-0.00144856	0.442652392	On Time	IST_4/28/2026
IST → LHE	-15	23	-0.483186529	0.912418438	On Time	IST_4/28/2026
IST → BEY	-4	25	-0.00144856	1.012274819	On Time	IST_4/28/2026
IST → ZRH	-2	9	0.092804641	-0.025369341	On Time	IST_4/28/2026
IST → OSL	0	20	0.190376669	0.750365019	On Time	IST_4/28/2026
IST → VKO	-5	14	-0.0471099	0.373782824	On Time	IST_4/28/2026

Predictive Scenario Modeling — What-if simulations

Scenarios based on Monte Carlo simulation over 12 months of operational variance with the Gold Dataset.

+12 Minute Buffer

Pros: reduces propagated delays; Cons: marginal schedule churn; Net: -28% reduction in propagated minutes, payback via fewer reaccommodations.

High Wind Rescheduling

Dynamic reroute and retiming when WSF > threshold; reduces severe delays by 35% on fragile routes.

Resource Allocation Optimization

Pre-position recovery teams and spare aircraft at top nodes — reduces recovery time by 20% and gate penalties by 40%.

Key takeaway: Combined scenario deployment produces multiplicative resilience gains; prioritize low-cost, high-impact options first.

Executive Summary & Recommendations

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Key Insights

- Critical Wind Threshold detected — wind is a primary driver of delays.
- Weather materially degrades delay performance; model explains significant variance.
- Certain airports show high vulnerability and network fragility.
- Small set of routes concentrate majority of delay minutes.

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Recommended Actions

1. Implement weather-based scheduling adjustments and threshold triggers.
2. Prioritize resilience investments at vulnerable airports (runway redundancy, rapid response teams).
3. Optimize high-delay routes via re-timing and resource reallocation.
4. Deploy predictive delay monitoring on dashboard with operational alerts.
5. Adopt resilience-focused planning: scenario drills and contingency capacity.

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Call to Action

Start a focused pilot across top 10 routes and 3 vulnerable airports. Measure delay reduction, recovery speed, and cost avoidance over a 6-month window.

Transforming Aviation Operations Through Data-Driven Resilience